

Multi-View Invariant Learning for Soccer Player Re-Identification

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Abstract

Player re-identification in broadcast soccer is a challenging fine-grained visual recognition problem because players from the same team often share nearly identical uniforms, appear under severe viewpoint changes, suffer from motion blur and occlusion, and are observed only through a small number of thumbnails per identity. This project studies whether identity-preserving representations can be learned under strong camera-view, pose, lighting, scale, and occlusion shifts. We propose a multi-view invariant player re-identification framework that combines global identity learning, local body-part-aware representation, view-invariance regularization, team-aware hard negative mining, occlusion-robust feature aggregation, and replay-action consistency. The project is designed around the SoccerNet Re-Identification benchmark, where each query player crop from an action frame must be matched against player crops from corresponding replay frames. The central research objective is to learn embeddings that are discriminative for identity while invariant to nuisance factors such as camera viewpoint, player pose, crop resolution, illumination, and partial occlusion.

1 Project Motivation

Soccer player re-identification differs from standard person re-identification. In surveillance-style ReID, clothing is often a useful identity cue. In soccer, this assumption breaks down because all players from the same team wear highly similar kits. Therefore, the model must rely on subtle identity-preserving cues such as body shape, jersey number when visible, local body-part appearance, pose-normalized texture, spatial context, and multi-view consistency.

The central question is:

Can a computer vision model learn identity-preserving representations for soccer players under severe camera-view, pose, lighting, scale, resolution, and occlusion shifts?

The broader computer vision contribution is:

Learning robust fine-grained identity representations when appearance is highly homogeneous within groups and nuisance variation across views is stronger than inter-identity variation.

2 Dataset

2.1 Primary Dataset: SoccerNet Re-Identification

The recommended dataset is **SoccerNet Re-Identification**. It contains player thumbnails extracted from broadcast soccer videos. The task is to re-identify players across multiple camera viewpoints depicting the same action. In the benchmark setting, a player crop from the action frame is used as a query, and crops from replay frames are used as the gallery. The objective is to rank gallery crops such that the same player identity appears as high as possible.

The dataset is suitable for project because it naturally contains:

- multi-view player observations,
- action-frame to replay-frame correspondences,
- strong camera-view changes,
- low-resolution and variable-resolution thumbnails,
- same-team visual ambiguity,
- occlusion and motion blur,
- few samples per identity.

2.2 Dataset Organization

A typical ReID dataset consists of:

$$\mathcal{D} = \{(x_i, y_i, c_i, m_i, v_i)\}_{i=1}^N, \quad (1)$$

where:

- x_i is the player crop,
- y_i is the player identity label,
- c_i is the camera/view label if available,
- m_i is the match identifier,
- v_i is the view type, e.g., action frame or replay frame.

For retrieval evaluation, each query image q has a gallery:

$$\mathcal{G}(q) = \{g_1, g_2, \dots, g_M\}. \quad (2)$$

The model produces an embedding:

$$z = f_\theta(x) \in \mathbb{R}^d. \quad (3)$$

Gallery images are ranked by distance or similarity:

$$s(q, g) = \frac{f_\theta(q)^\top f_\theta(g)}{\|f_\theta(q)\|_2 \|f_\theta(g)\|_2}. \quad (4)$$

The correct player should have the highest similarity.

3 Research Objective

The primary objective is to learn an embedding function:

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d \quad (5)$$

such that:

$$\|f_\theta(x_i) - f_\theta(x_j)\|_2 \quad (6)$$

is small when x_i and x_j show the same player, even under different viewpoints, and large when they show different players, especially different players from the same team.

Formally:

$$d(f_\theta(x_i), f_\theta(x_j)) \rightarrow 0 \quad \text{if } y_i = y_j, \quad (7)$$

and:

$$d(f_\theta(x_i), f_\theta(x_j)) \rightarrow \infty \quad \text{if } y_i \neq y_j. \quad (8)$$

However, for soccer ReID, the hard case is:

$$y_i \neq y_j \quad \text{and} \quad \text{team}(x_i) = \text{team}(x_j), \quad (9)$$

because same-team players have highly similar clothing.

Therefore, the project should optimize not only general identity separation but also **same-team identity discrimination**.

4 Core Research Questions

4.1 RQ1: Multi-View Invariance

Can the model learn identity embeddings that are invariant to camera-view changes?

$$f_\theta(x_i^{\text{view}_1}) \approx f_\theta(x_i^{\text{view}_2}) \quad \text{for the same player } i. \quad (10)$$

4.2 RQ2: Same-Team Discrimination

Can the model separate two players from the same team who wear the same kit?

$$f_\theta(x_i) \not\approx f_\theta(x_j) \quad \text{when } y_i \neq y_j \text{ but } \text{team}_i = \text{team}_j. \quad (11)$$

4.3 RQ3: Occlusion Robustness

Can the model maintain identity consistency when only part of the body is visible?

$$f_\theta(x_i) \approx f_\theta(\text{Mask}(x_i)). \quad (12)$$

4.4 RQ4: Local-Part Identity Cues

Do local regions such as head, torso, legs, jersey number, and body silhouette improve player identity representation?

4.5 RQ5: Replay-Action Consistency

Can replay frames be used as natural multi-view positives for robust self-supervised or supervised contrastive learning?

5 Proposed Method: MV-SoccerReID

We define the proposed framework as:

MV-SoccerReID: Multi-View Invariant Soccer Player Re-Identification

The method contains five major components:

1. global identity encoder,
2. local body-part-aware branch,
3. multi-view invariant contrastive learning,
4. same-team hard negative mining,
5. occlusion-robust feature aggregation.

6 Model Architecture

6.1 Input

Each player crop is resized to a fixed resolution:

$$x_i \in \mathbb{R}^{H \times W \times 3}. \quad (13)$$

Common choices are:

$$H \times W \in \{256 \times 128, 224 \times 224, 384 \times 192\}. \quad (14)$$

For transformer-based models, the image is split into patches:

$$x_i \rightarrow \{p_1, p_2, \dots, p_P\}. \quad (15)$$

Each patch is embedded as:

$$e_p = W_p p + b_p. \quad (16)$$

6.2 Backbone Encoder

Use either a convolutional or transformer-based backbone.

Recommended backbones:

- ResNet-50,
- OSNet,
- Swin Transformer,
- ViT-Small,
- ConvNeXt-Tiny.

For a this project, a transformer or hybrid CNN-transformer model is preferable.

The backbone produces a feature map:

$$F_i = B_\theta(x_i) \in \mathbb{R}^{C \times H' \times W'}. \quad (17)$$

Global average pooling gives:

$$z_i^g = \text{GAP}(F_i). \quad (18)$$

The global embedding is:

$$e_i^g = \frac{W_g z_i^g}{\|W_g z_i^g\|_2}. \quad (19)$$

6.3 Local Body-Part Branch

Because same-team players have similar uniforms, global appearance may be insufficient. Therefore, the model should learn local body-part representations.

Divide the feature map into K horizontal stripes:

$$F_i = [F_i^1, F_i^2, \dots, F_i^K]. \quad (20)$$

Each part feature is:

$$z_i^k = \text{GAP}(F_i^k), \quad (21)$$

and projected as:

$$e_i^k = \frac{W_k z_i^k}{\|W_k z_i^k\|_2}. \quad (22)$$

The local representation is:

$$e_i^{local} = [e_i^1; e_i^2; \dots; e_i^K]. \quad (23)$$

The final embedding is:

$$e_i = \text{Normalize} \left(W_f [e_i^g; e_i^{local}] \right). \quad (24)$$

6.4 Pose-Guided Part Attention

Horizontal stripes are simple but may be noisy under pose changes. A stronger model uses part attention maps.

Let:

$$A_i^k \in [0, 1]^{H' \times W'} \quad (25)$$

be the attention map for body part k , such as head, torso, legs, and feet.

The part representation is:

$$z_i^k = \frac{\sum_{u,v} A_i^k(u, v) F_i(u, v)}{\sum_{u,v} A_i^k(u, v) + \epsilon}. \quad (26)$$

The part maps may be obtained by:

- a lightweight learned attention module,
- pretrained human pose estimation,
- unsupervised part discovery,
- transformer attention tokens.

For an implementable first version, use horizontal part pooling.

7 Loss Functions

The final training objective should combine several losses:

$$\mathcal{L} = \mathcal{L}_{id} + \lambda_1 \mathcal{L}_{tri} + \lambda_2 \mathcal{L}_{mv} + \lambda_3 \mathcal{L}_{team} + \lambda_4 \mathcal{L}_{occ} + \lambda_5 \mathcal{L}_{view}. \quad (27)$$

Each term is explained below.

7.1 Identity Classification Loss

Each embedding is classified into player identities:

$$\hat{y}_i = \text{softmax}(W_c e_i + b_c). \quad (28)$$

The identity loss is:

$$\mathcal{L}_{id} = - \sum_i \log p_\theta(y_i | x_i). \quad (29)$$

This encourages identity-level discrimination.

7.2 Triplet Loss

For an anchor x_a , positive x_p , and negative x_n :

$$y_a = y_p, \quad y_a \neq y_n. \quad (30)$$

The triplet loss is:

$$\mathcal{L}_{tri} = \sum [m + d(e_a, e_p) - d(e_a, e_n)]_+, \quad (31)$$

where m is the margin and:

$$d(e_i, e_j) = \|e_i - e_j\|_2. \quad (32)$$

This directly enforces metric separation.

7.3 Supervised Contrastive Loss

For a batch of embeddings, define positives:

$$P(i) = \{p : y_p = y_i, p \neq i\}. \quad (33)$$

The supervised contrastive loss is:

$$\mathcal{L}_{supcon} = - \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(e_i^\top e_p / \tau)}{\sum_{a \neq i} \exp(e_i^\top e_a / \tau)}. \quad (34)$$

This is particularly useful when multiple views of the same player are available.

7.4 Multi-View Invariance Loss

Let x_i^a be the action-frame crop and x_i^r be a replay-frame crop of the same player.

The multi-view invariance loss is:

$$\mathcal{L}_{mv} = \sum_i \|f_\theta(x_i^a) - f_\theta(x_i^r)\|_2^2. \quad (35)$$

Alternatively, use contrastive multi-view learning:

$$\mathcal{L}_{mv-con} = - \sum_i \log \frac{\exp(s(e_i^a, e_i^r) / \tau)}{\sum_j \exp(s(e_i^a, e_j^r) / \tau)}. \quad (36)$$

This explicitly pulls together action-frame and replay-frame views of the same identity.

7.5 Team-Aware Hard Negative Loss

Same-team players are the hardest negatives. Define:

$$N_{team}(i) = \{j : y_j \neq y_i, team_j = team_i\}. \quad (37)$$

The team-aware hard negative loss is:

$$\mathcal{L}_{team} = - \sum_i \log \frac{\exp(e_i^\top e_{p(i)}/\tau)}{\exp(e_i^\top e_{p(i)}/\tau) + \sum_{j \in N_{team}(i)} \exp(e_i^\top e_j/\tau)}. \quad (38)$$

This forces the model to distinguish visually similar players wearing the same uniform.

7.6 View-Adversarial Loss

The embedding should not encode camera viewpoint strongly. Let v_i be the view label, such as action frame or replay frame, or camera cluster if available.

A view classifier predicts:

$$\hat{v}_i = g_\psi(e_i). \quad (39)$$

The view classification loss is:

$$\mathcal{L}_{view-cls} = - \sum_i \log p_\psi(v_i | e_i). \quad (40)$$

Using gradient reversal, train the encoder to remove view information:

$$\min_{\theta} \max_{\psi} \mathcal{L}_{id} - \lambda_{view} \mathcal{L}_{view-cls}. \quad (41)$$

This encourages view-invariant identity features.

7.7 Occlusion Consistency Loss

Apply random erasing or patch masking:

$$\tilde{x}_i = \text{Mask}(x_i). \quad (42)$$

Then enforce:

$$\mathcal{L}_{occ} = \sum_i \|f_\theta(x_i) - f_\theta(\tilde{x}_i)\|_2^2. \quad (43)$$

This teaches the model to maintain identity consistency under partial visibility.

7.8 Part Consistency Loss

For local part embeddings, enforce consistency across views:

$$\mathcal{L}_{part} = \sum_i \sum_{k=1}^K \|e_{i,a}^k - e_{i,r}^k\|_2^2. \quad (44)$$

This is useful when corresponding body regions remain visible across action and replay frames.

8 Training Strategy

8.1 Batch Construction

A standard random batch may not contain enough positives or hard negatives. Use identity-balanced sampling.

Let:

$$B = P \times K \quad (45)$$

where:

- P identities are sampled,
- K images per identity are sampled.

This enables triplet and supervised contrastive learning.

8.2 Team-Aware Sampling

For each batch, sample identities from the same team whenever possible:

$$\mathcal{B} = \{x_i : team_i = t\} \cup \{x_j : team_j \neq t\}. \quad (46)$$

This creates hard negatives naturally.

8.3 View-Paired Sampling

If action-replay pairs are available, sample paired views:

$$(x_i^{action}, x_i^{replay}). \quad (47)$$

These pairs are ideal positives for multi-view invariance learning.

9 Algorithm

10 Evaluation Protocol

10.1 Retrieval Setup

For each query q , compute its embedding:

$$e_q = f_\theta(q). \quad (48)$$

For every gallery image g_j :

$$e_j = f_\theta(g_j). \quad (49)$$

Rank gallery images by cosine similarity:

$$s(q, g_j) = e_q^\top e_j. \quad (50)$$

Algorithm 1 Training MV-SoccerReID

- 1: Input: training set $\mathcal{D}$, identity labels y , team labels t , view labels v
 - 2: Initialize encoder f_θ , classifier C , view discriminator g_ψ
 - 3: **for** each training iteration **do**
 - 4: Sample identity-balanced mini-batch $\mathcal{B}$
 - 5: Include same-team hard negatives where possible
 - 6: Include action-replay positive pairs where available
 - 7: Apply augmentations and random occlusion masks
 - 8: Compute embeddings $e_i = f_\theta(x_i)$
 - 9: Compute identity loss $\mathcal{L}_{id}$
 - 10: Compute triplet or supervised contrastive loss $\mathcal{L}_{tri}$ or $\mathcal{L}_{supcon}$
 - 11: Compute multi-view invariance loss $\mathcal{L}_{mv}$
 - 12: Compute team-aware hard negative loss $\mathcal{L}_{team}$
 - 13: Compute occlusion consistency loss $\mathcal{L}_{occ}$
 - 14: Compute view-adversarial loss $\mathcal{L}_{view}$
 - 15: Update θ and ψ using the total objective
 - 16: **end for**
 - 17: Return trained encoder f_θ
-

10.2 Rank-k Accuracy

Rank- k accuracy is:

$$\text{Rank@k} = \frac{1}{Q} \sum_{q=1}^Q \mathbb{I}[\text{correct identity appears in top } k]. \quad (51)$$

Report:

$$\text{Rank@1}, \quad \text{Rank@5}, \quad \text{Rank@10}. \quad (52)$$

10.3 Mean Average Precision

For each query q , average precision is:

$$AP(q) = \frac{1}{R_q} \sum_{k=1}^M P_q(k) \text{rel}_q(k), \quad (53)$$

where:

- R_q is the number of relevant gallery images,
- $P_q(k)$ is precision at rank k ,
- $\text{rel}_q(k) = 1$ if the item at rank k is relevant.

Mean average precision is:

$$mAP = \frac{1}{Q} \sum_{q=1}^Q AP(q). \quad (54)$$

10.4 Same-Team Retrieval Accuracy

Because same-team players are hardest, evaluate separately on gallery candidates from the same team:

$$\text{Rank@1}_{\text{same-team}} = \frac{1}{Q} \sum_{q=1}^Q \mathbb{I}[\arg \max_{g \in \mathcal{G}_{\text{same-team}}(q)} s(q, g) \text{ is correct}]. \quad (55)$$

This is an important metric for soccer ReID.

10.5 Cross-View Retrieval Accuracy

Evaluate action-to-replay and replay-to-action retrieval separately:

$$\text{Rank@1}_{A \rightarrow R}, \quad \text{Rank@1}_{R \rightarrow A}. \quad (56)$$

This directly measures multi-view invariance.

10.6 Occlusion Robustness

Create masked query images:

$$\tilde{q} = \text{Mask}(q). \quad (57)$$

Evaluate:

$$\Delta mAP_{\text{occ}} = mAP(q) - mAP(\tilde{q}). \quad (58)$$

Lower degradation means stronger occlusion robustness.

10.7 Resolution Robustness

Downsample query images:

$$q^{\text{low}} = \text{Downsample}(q). \quad (59)$$

Evaluate:

$$\Delta mAP_{\text{res}} = mAP(q) - mAP(q^{\text{low}}). \quad (60)$$

11 Baselines

Use strong baselines to make the project research-grade.

11.1 Backbone Baselines

1. ResNet-50 + ID loss,
2. ResNet-50 + ID + Triplet,
3. OSNet,
4. ViT-Small,
5. Swin Transformer,
6. ConvNeXt-Tiny.

11.2 Loss Baselines

1. cross-entropy only,
2. triplet loss,
3. supervised contrastive loss,
4. centroid loss,
5. circle loss,
6. ArcFace or CosFace margin loss.

11.3 Ablation Baselines

1. without local branch,
2. without multi-view invariance,
3. without team-aware negatives,
4. without view-adversarial learning,
5. without occlusion consistency,
6. without part consistency,
7. without identity-balanced sampling.

12 Main Experiments

12.1 Experiment 1: Standard ReID Benchmark

Compare the proposed method against baselines on:

$$mAP, \quad Rank@1, \quad Rank@5, \quad Rank@10. \quad (61)$$

Expected table:

Table 1: Standard player re-identification results.

Method	mAP	Rank@1	Rank@5	Rank@10
ResNet-50 + CE	—	—	—	—
ResNet-50 + CE + Triplet	—	—	—	—
OSNet	—	—	—	—
ViT-Small	—	—	—	—
Swin Transformer	—	—	—	—
MV-SoccerReID	—	—	—	—

12.2 Experiment 2: Same-Team Confusion Analysis

Evaluate retrieval only among players from the same team.

This experiment directly tests whether the method solves the most important soccer-specific ambiguity.

Table 2: Same-team retrieval performance.

Method	Same-Team mAP	Same-Team Rank@1	Same-Team Rank@5
Baseline	—	—	—
+ Local Branch	—	—	—
+ Team-Aware Loss	—	—	—
Full Method	—	—	—

12.3 Experiment 3: Multi-View Invariance

Evaluate action-to-replay and replay-to-action retrieval.

Table 3: Cross-view retrieval performance.

Method	$A \rightarrow R$ mAP	$A \rightarrow R$ R1	$R \rightarrow A$ mAP	$R \rightarrow A$ R1
Baseline	—	—	—	—
+ Multi-View Loss	—	—	—	—
+ View-Adversarial	—	—	—	—
Full Method	—	—	—	—

12.4 Experiment 4: Occlusion Robustness

Evaluate performance under artificial occlusion.

Table 4: Occlusion robustness.

Method	No Mask mAP	Masked mAP	Drop
Baseline	—	—	—
+ Random Erasing	—	—	—
+ Occlusion Consistency	—	—	—
Full Method	—	—	—

12.5 Experiment 5: Part Branch Ablation

Test how many body parts are useful.

12.6 Experiment 6: Embedding Visualization

Use t-SNE or UMAP to visualize embeddings.

Expected visualizations:

- same identity across views should cluster together,
- same-team different identities should separate,
- action and replay views should overlap in feature space,
- occluded and clean crops should remain close.

Table 5: Effect of part decomposition.

Parts	mAP	Rank@1	Same-Team Rank@1
Global only	–	–	–
2 parts	–	–	–
4 parts	–	–	–
6 parts	–	–	–
Learned part attention	–	–	–

13 Research-Level Novelty

The method should not be presented as just another ReID model. The novelty should be framed around the following contributions.

13.1 Contribution 1: Soccer-Specific ReID Challenge

Soccer ReID is uniquely difficult because clothing is not discriminative within a team. This creates a fine-grained identity learning problem under homogeneous appearance.

13.2 Contribution 2: Multi-View Invariant Identity Learning

The model explicitly uses action-replay correspondences to learn identity-preserving representations across viewpoint changes.

13.3 Contribution 3: Same-Team Hard Negative Learning

The method introduces a loss or sampling strategy that prioritizes visually similar same-team negatives.

13.4 Contribution 4: Occlusion-Robust Part-Aware Representation

The model combines global and local body-part features, making it robust to partial visibility.

13.5 Contribution 5: Detailed Robustness Evaluation

The paper evaluates not only standard mAP and Rank@1 but also same-team retrieval, cross-view retrieval, occlusion robustness, and resolution robustness.

14 Implementation Backbone

14.1 Recommended Repository Structure

MV-SoccerReID/

```

configs/
  baseline_resnet50.yaml
  osnet.yaml
  vit_small.yaml
  mv_soccerreid.yaml

data/
  download_soccernet.py
  build_metadata.py

```

```

split_dataset.py
datasets.py

models/
  backbones/
    resnet.py
    osnet.py
    vit.py
    swin.py
  part_pooling.py
  attention_parts.py
  view_discriminator.py
  mv_soccerreid.py

losses/
  id_loss.py
  triplet_loss.py
  supcon_loss.py
  multiview_loss.py
  team_aware_loss.py
  occlusion_loss.py
  adversarial_loss.py

samplers/
  identity_sampler.py
  team_aware_sampler.py
  view_pair_sampler.py

train.py
evaluate.py
extract_features.py
visualize_embeddings.py
robustness_eval.py
README.md

```

14.2 Data Loader Output

Each training sample should return:

```

{
  "image": image_tensor,
  "pid": player_identity,
  "team_id": team_identity,
  "match_id": match_identity,
  "view_id": action_or_replay,
  "camera_id": optional_camera_label,
  "bbox": bounding_box,
  "image_path": file_path
}

```

14.3 Training Configuration

Recommended starting configuration:

```
image_size: [256, 128]
backbone: resnet50
embedding_dim: 512
num_parts: 6
batch_p: 16
batch_k: 4
batch_size: 64
optimizer: adamw
learning_rate: 3e-4
weight_decay: 1e-4
epochs: 80
warmup_epochs: 5
margin: 0.3
temperature: 0.07
lambda_triplet: 1.0
lambda_multiview: 0.5
lambda_team: 0.5
lambda_occ: 0.2
lambda_view: 0.1
```

14.4 Data Augmentation

Use:

- random horizontal flip,
- color jitter,
- random erasing,
- Gaussian blur,
- random grayscale,
- random crop and resize,
- resolution degradation,
- motion blur simulation.

The augmentation distribution should simulate broadcast video challenges.

15 Failure Cases to Analyze

Important failure categories:

1. same-team players with no visible jersey number,
2. severe occlusion,
3. tiny or blurred crops,

4. goalkeeper versus field player ambiguity,
5. incorrect bounding boxes,
6. replay color/lighting shift,
7. pose mismatch between action and replay views,
8. back-view versus front-view identity ambiguity.

16 Expected Figures

16.1 Figure 1: Overall Architecture

Show:

Input crop $\rightarrow$ Backbone $\rightarrow$ Global branch + Part branch $\rightarrow$ Embedding $\rightarrow$ ReID losses.

Also show:

Action frame $\leftrightarrow$ Replay frame

as multi-view positives.

16.2 Figure 2: Same-Team Hard Negatives

Show one query player, one positive replay crop, easy negative from another team, and hard negative from the same team.

16.3 Figure 3: Part-Aware Representation

Show horizontal part regions or learned attention maps over body regions.

16.4 Figure 4: Embedding Space

Use t-SNE or UMAP to show that the proposed method clusters the same player across views while separating same-team players.

16.5 Figure 5: Robustness

Show retrieval under clean, occluded, low-resolution, and blurred inputs.